

Final Report: sVAD Reconstruction

We begin the reconstruction of the model by establishing already defined parameters. We notice that the given structure is a diagonal network. In a recurrent diagonal network the feedback is only based on its own previous state. The weight matrix is thus a diagonal matrix, as opposed to a dense network where feedback is based on all previous states, thus make an enriched matrix.

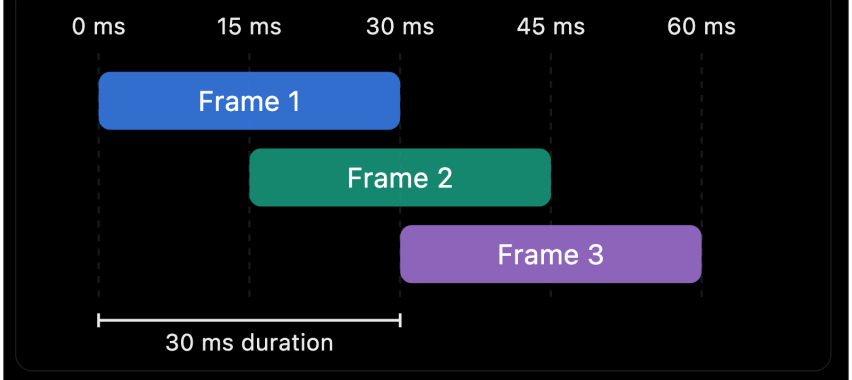
Parameter	Value
Frame duration	30 ms
Frame overlap	50%
Frame shift	15 ms
Encoded features	20
Recurrent hidden neurons	32
Output neurons	2
LIF decay constant **α**	0.5
Firing threshold **θ**	0.3
Surrogate gradient parameter **a**	4
Batch size	128
Training epochs	100
Optimizer	Adam
Initial learning rate	0.001
LR schedule	Multiply by 0.1 every 40 epochs

Let us define some terms:

▼ Frame Duration, frame overlap and frame shift

The paper uses:

- Frame duration = 30 ms
- Frame overlap = 50%
- Frame shift = 15 ms



Frame duration: How much audio is contained in each window. Here, each frame covers 30 ms.

Frame overlap: How much audio is shared between consecutive windows. A 50% overlap means consecutive 30 ms frames share 15 ms of audio.

Frame shift: How far we move forward before extracting the next frame.

consecutive 30 ms frames share 15 ms of audio.

Frame shift: How far we move forward before extracting the next frame.

$$\text{Frame shift} = 30(1 - 0.5) = 15 \text{ ms}$$

So we make a new speech/non-speech decision every 15 ms.

At a sample rate of 16 kHz, for example, one 30 ms frame contains 480 audio samples and the shift is 240 samples. The 16 kHz rate is an illustrative assumption, not a sampling rate explicitly specified in this paper's experimental setup.

▼ Encoded Features

Raw audio is just a sequence of amplitude values.

The auditory encoder transforms this waveform into more useful representations, such as learned responses to different frequency bands.

In sVAD, the encoder produces 20-dimensional features.

$$S[t] = [s_1, s_2, \ldots, s_{20}]$$

▼ Recurrent hidden neurons

The paper uses 32 recurrent spiking neurons in its classifier. These are neurons that maintain internal membrane states and have recurrent connections. These constitute the hidden layers.

▼ Output Neurons

2 linear readout neurons → [speech, not speech]. Then softmax function applied to create it into a probability. This then is used to train the module using cross entropy.

$$p_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

▼ Neural Network Constants (α, a)

$$U[t] = \alpha U[t-1] + I[t] - \vartheta O[t-1]$$

For the equation for Leaky Integrate and Fire NN

LIF Decay Constant (α): The decay constant α determines how much previous membrane potential is retained. A smaller α means the neuron forgets its previous voltage more quickly. A value closer to 1 means it retains it longer.

Surrogate Gradient Parameter (a): The spike function is a step function. Its true derivative is zero almost everywhere and undefined at the threshold, making ordinary backpropagation ineffective. Instead of using the actual derivative, it approximates it by:

$$\frac{\partial O}{\partial U} \approx \begin{cases} \frac{1}{a}, & |U - \vartheta| < \frac{a}{2} \\ 0, & \text{otherwise} \end{cases}$$

During the forward pass, the neuron still produces hard 0/1 spikes. The surrogate gradient is used only for calculating gradients during backpropagation.

▼ Everything else

Term	What it means	Paper's setting
Batch size	Number of training samples processed before each optimizer update. A sample may contain a sequence of frames.	128
Training epochs	Number of complete passes through the training dataset.	100
Optimizer	Algorithm that updates weights using computed gradients.	Adam
Initial learning rate	Initial scale of the optimizer's parameter updates.	0.001
LR schedule	Rule that changes the learning rate during training.	Reduce by a factor of 10 every 40 epochs

However, it is important to note that a lot of parameters that were essential to the project's reconstruction were not given, such as

1. The Number of SNN Timesteps (How long the SNN operates before making a prediction)
2. Exact SincNet Configuration (Band pass filters constructed to allow the network to learn which frequency regions are useful for detecting speech)
3. sConv1D parameters (To extract more complex auditory patterns and convert the representation into spikes)
4. Attention-layer structure (To make a decision to consider which auditory features are to be considered and which to not)
5. Loss-weighting hyperparameter (λ) (It balances speech classification accuracy against the reconstruction of clean speech features from noisy inputs.)

The expected results are as follows:

Noise condition	Target HTER
Low noise	4.0%
Medium noise	11.9%
High noise	19.1%

The project was divided into 8 code-based stages.

Stage 1

It entailed the general reconstruction of the architecture proposed in the ICASSP 2024 Paper, with the reported parameter count and experimental results as reference for our assumptions.

Our initial PyTorch reconstruction contains:

- Auditory encoder: SincNet followed by spiking 1D convolution (sConv1D).
- Attention mechanism: Three spiking fully connected layers generating a feature mask.
- Classifier: Spiking recurrent neural network (sRNN) with 32 hidden neurons and 2 output neurons.
- Training: Cross-entropy classification loss combined with MSE attention loss.

The configuration details are as follows. The assumed values are clearly stated.

Parameter	Value
Sample rate	16 kHz (assumed)
Frame duration / shift	30 ms / 15 ms
Encoded features	20
SincNet kernel size	101 (assumed)
sConv1D kernel size	3
Attention hidden widths	31, 31
Recurrent neurons	32
Recurrence type	Diagonal
LIF decay constant **α**	0.5
Firing threshold **θ**	0.3
Surrogate gradient parameter **a**	4
Loss weight **λ**	1.0 (assumed)

Our results were as follows:

Model	Parameters
Our reconstruction	4313
Paper’s repoted count	4300

Our reconstruction included:

Component	Parameters
Encoder	1,260
Attention	2,283
Classifier	770
Total	4,313

The architecture search identified 1,214 configurations within ±100 parameters of the reported 4,300.

Several configurations contain exactly 4,300 parameters, demonstrating that parameter count alone cannot uniquely identify the original model.

Component	Function	Why it has a minimum parameter requirement
Encoder (1,260)	Extracts speech-relevant features from raw audio using SincNet and spiking convolution.	Needs sufficient filters and connections to generate the required 20-dimensional feature representation.
Attention (2,283)	Selectively preserves useful encoded features while suppressing noise-corrupted information.	Needs enough neurons and connections to generate a 20-dimensional attention mask from the encoded features.
Classifier (770)	Uses a spiking recurrent network to identify speech and non-speech frames.	Requires connections between 20 input features, 32 recurrent hidden neurons, and 2 output neurons.

The numbers selected were conservative approximations, and were trained upon in further stages.

The implementation yielded the following losses:

$$L_{\text{CE}} = 0.62549, \qquad L_{\text{MSE}} = 0.42$$

$$Therefore,\ L_{\text{total}} = 1.04549$$

where, **CE (Cross-Entropy Loss):** Measures how accurately the model predicts speech versus non-speech. Lower CE means better classification predictions.

MSE (Mean Squared Error): Measures the difference between the clean reference features and the noisy features after attention. Lower MSE means the two feature representations are more similar.

The next objective was to move from parameter-count matching to experimental validation, using speech-detection accuracy and noise robustness to distinguish plausible architectures.

Stage 2

In stage 2, we verified that our initial PyTorch reconstruction could process labelled audio, learn speech/non-speech distinctions, and perform correct frame-level classification.

We used synthetic tone-and-silence recordings before introducing real speech data from **Google Speech Commands v0.02.** This allowed us to debug the architecture, training process, and evaluation pipeline under controlled conditions.

We defined Half Time Error Rate as:

$$Half\ Time\ Error\ Rate\ (HTER) = \frac{False\ Alarm\ Rate + Miss\ Rate}{2}$$

Our experiments yielded:

Experiment	Best HTER	Final HTER
Unweighted CE, λ=1	44.3%	50.0%
Unweighted CE, λ=0	43.8%	50.0%
Weighted CE, λ=1	45.1%	54.9%
Weighted CE, λ=0	**21.9%**	**21.9%**

📝 The first two experiments demonstrated a classification collapse: the network eventually predicted almost every frame as speech.
Removing the MSE loss alone did not resolve this issue.

Inspection of the sampled training data revealed approximately 74.7% speech frames and 25.3% non-speech frames.

We introduced class-weighted cross-entropy, initially using weights of approximately 1.98 for non-speech and 0.67 for speech.

Using weighted cross-entropy and λ=0, we extended training to 30 epochs.

The training (or total) loss decreased from **0.7014** at epoch 1 to **0.2402** at epoch 30.

The best validation performance occurred at epoch 20:

- **MR (Miss Rate): 9.5%**
- **FAR (False Alarm Rate): 0.0%**
- **HTER (Half-Total Error Rate): 4.8%**

The final epoch produced an HTER of 10.3%.

The results were as follows:

Metric	Result
Initial training HTER	48.89%
Final training HTER	**1.39%**
Final cross-entropy loss	0.0313
True positives	40
False negatives	0
False positives	1
True negatives	35
Training accuracy	**98.68%**

The model correctly classified 75 of 76 frames.

However, the precise SincNet configuration, spiking convolution parameters, attention structure, recurrent connectivity, internal SNN timesteps, and original loss weight remain unresolved.

Stage 3 will focus on real speech recordings, clean/noisy audio pairs, noise robustness, fixed validation protocols, and evaluating architectural candidates against meaningful voice activity detection benchmarks.

Stage 3

Moving from Stage 2's synthetic tone-and-silence debugging task to real spoken-word recordings, we measured frame-level speech/non-speech detection under multiple signal-to-noise ratios (SNRs), and inspected the model's temporal errors visually.

The dataset used is as follows:

- **Dataset:** A pilot subset of Google Speech Commands v0.02, prepared with **180 training, 50 validation, and 50 test recordings** (as configured in the preparation command).
- **Audio:** 16 kHz; 30 ms analysis frames with 15 ms hop (50% overlap); model outputs a binary speech decision per frame.
- **Annotations:** Frame-level speech intervals were estimated from signal energy during preparation. They are **pseudo-labels**, not manually annotated VAD ground truth. HTER therefore measures agreement with these provisional labels.
- **Noise:** Controlled additive noise, described in the pilot as Gaussian; not the original paper's recorded QUT-NOISE-TIMIT environments.
- **Evaluation:** HTER = (MR + FAR)/2, where MR = FN/(TP+FN) and FAR = FP/(FP+TN). Lower is better. Six test conditions: −10, −5, 0, +5, +10, +15 dB SNR.
- **Test volume:** Across each SNR condition, evaluation reported **1,654 labelled speech frames and 3,296 labelled non-speech frames** (4,950 total).

However, admittedly, the experiments are a real-speech pilot, not a faithful reproduction of the original paper’s dataset, acoustic conditions, or annotation protocol.

The pilot used class-weighted cross-entropy and **λ = 0** (attention MSE disabled in the loss). The reported initial training counts were 11,195 non-speech and 6,625 speech frames, giving class weights approximately **[0.7959,**

1.3449] for [non-speech, speech]. Training ran on the Mac's **MPS** backend. Checkpointing selected the best validation HTER.

We divided Stage 3 into 4 Experiments

1. **Experiment A:** We trained the model with noise at 0dB SNR for 20 epochs. The results are as follows, and best validation HTER of 13.4% was achieved at epoch 20.

SNR (dB)	MR (%)	FAR (%)	HTER (%)
−10	35.31	7.01	21.16
−5	29.69	3.43	16.56
0	22.73	4.70	**13.72**
+5	16.87	14.75	15.81
+10	11.61	30.19	20.90
+15	7.74	50.39	29.07

   Observation: The model's false-alarm rate rose steeply at high SNR, while miss rate fell. This suggests sensitivity to the training noise distribution; it is not proof of a specific internal mechanism.

2. **Experiment B:** The same experiment was then conducted on multiple SNRs, retaining architecture and hyperparameter value. The results were as follows. There is general improvement across all but -10dB in HTER values, with notable improvements at higher SNR values.

SNR (dB)	MR (%)	FAR (%)	HTER (%)
−10	39.36	3.94	21.65
−5	31.08	4.28	17.68
0	26.12	3.73	14.93
+5	21.58	6.07	13.83
+10	18.50	9.50	14.00
+15	16.75	13.80	15.28

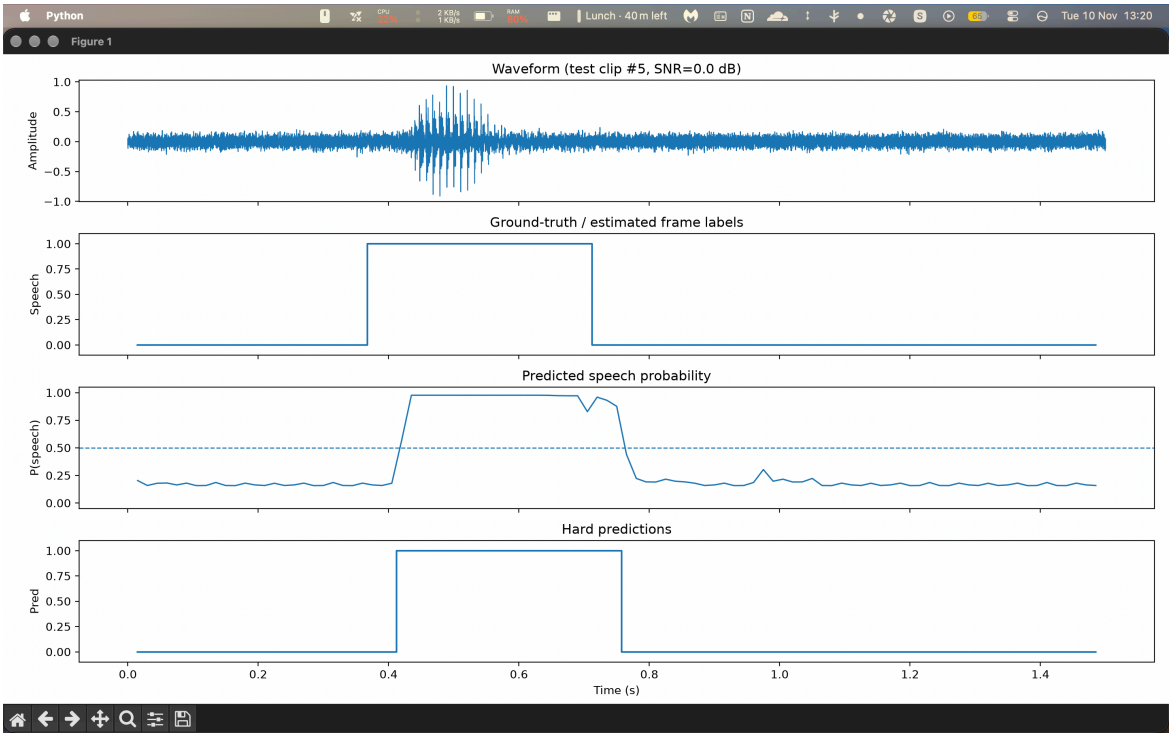
3. **Experiment C:** We ran the same experiment for 60 epochs. This gave us better results all across the board. The six-SNR mean HTER decreased from **16.23% (20-epoch multi-SNR)** to **14.55% (60-epoch multi-SNR)**

SNR (dB)	MR (%)	FAR (%)	HTER (%)
−10	33.37	7.37	20.37
−5	29.75	4.00	16.88
0	25.33	3.46	14.40
+5	22.31	2.97	12.64
+10	19.29	3.64	**11.46**
+15	17.29	5.76	11.53

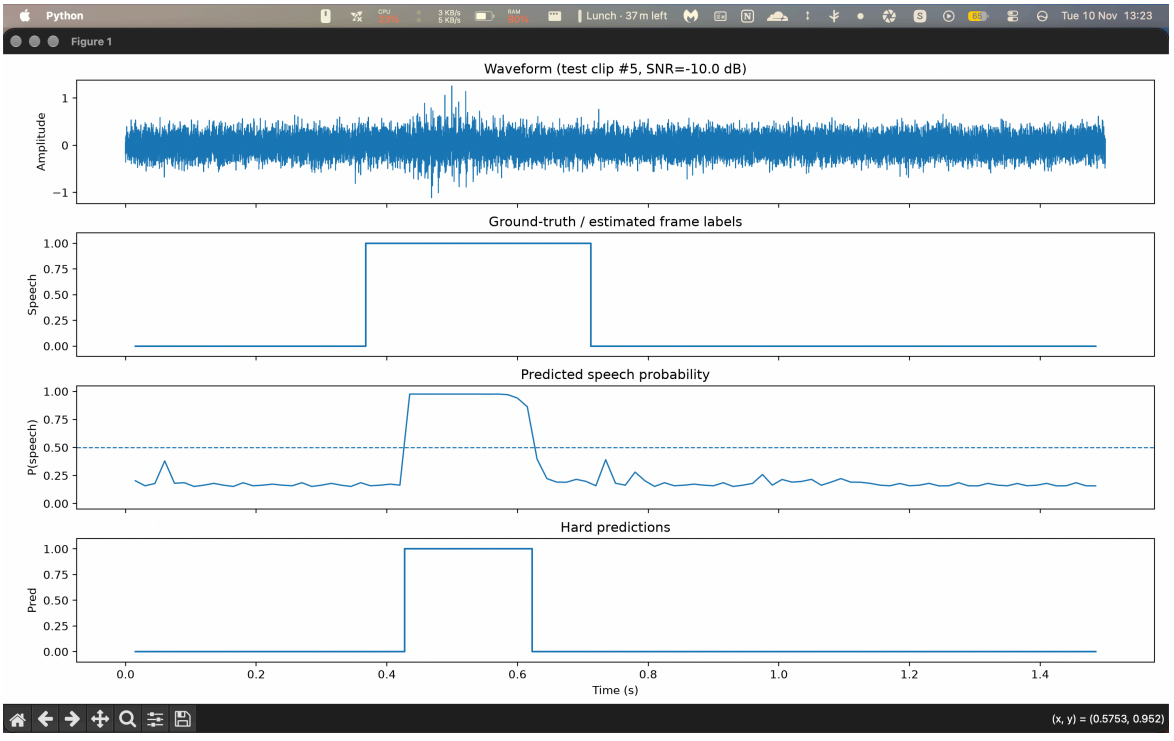
4. **Experiment D:** A visualization script was added to display, for a chosen test clip and SNR: the noisy waveform, energy-derived speech and non-speech labels, predicted speech probability with 0.5 decision threshold and binary predicted speech regions.

   On a representative clip at 0 dB, speech probability rose to approximately 0.95–0.98 during the estimated speech segment and stayed around 0.15–0.20 outside it. Hard predictions broadly matched the annotated speech region. Across additional manual visual checks, however, a **consistent apparent onset delay** was observed, with **offset behavior varying by noise condition**.

   Prediction for test clip 5, Noise = 0dB



Prediction for test clip 5, Noise = -10dB



Observation	Interpretation	Results
Predictions closely follow labels	Successful detection	Eh
Prediction begins late	Possible onset delay	Yes
Prediction ends late	Possible offset delay	Dependent on noise level
Speech is missed entirely	False negatives	No
Noise is classified as speech	False positives	No

Observation	Interpretation	Results
Prediction flickers rapidly	Temporal instability	No

Stage 4

Investigate whether the reconstructed sVAD classifier should use diagonal/self-recurrence or dense/full recurrence, while assessing model-size trade-offs and reproducibility. This stage does **not** establish the authors' original recurrence implementation.

The architectures used to compare were:

Component	Diagonal baseline	Full-size dense	Parameter-matched dense
Encoder parameters	1,260	1,260	1,260
Attention hidden widths	31, 31	31, 31	17, 24
Attention parameters	2,283	2,283	1,289
Classifier parameters	770	1,762	1,762
Total parameters	**4,313**	**5,305**	**4,311**

The dense recurrent matrix uses 32 × 32 = 1,024 weights, versus 32 self-recurrent weights for diagonal recurrence, accounting for 992 extra recurrent parameters. The matched dense architecture reduces attention width to offset those extra recurrent weights. Therefore the matched comparison changes *both* recurrence structure and attention capacity.

An experiment was ran on one seed (#42), followed by three seed repeatability over (#42, #123 and #2026)

The results of the first was as follows:

SNR	Diagonal 4,313	Full-size dense 5,305	Matched dense 4,311
−10 dB	20.37%	20.21%	**19.14%**
−5 dB	16.88%	16.39%	**16.20%**
0 dB	14.40%	14.88%	**13.73%**
+5 dB	12.64%	13.90%	**11.78%**
+10 dB	11.46%	13.28%	**11.09%**
+15 dB	**11.53%**	13.24%	11.56%
Six-SNR mean	**14.55%**	**15.32%**	**13.92%**

This suggested that the larger dense model did not outperform the diagonal model overall. However, the parameter matched dense configuration did outperform it overall, motivating replication. We thus discard the full size dense model, and compare the diagonal and matched dense models.

The second experiment yielded the following results:

Seed	Diagonal (4,313)	Matched dense (4,311)	Lower HTER
42	14.55%	**13.92%**	Matched dense
123	**14.33%**	14.67%	Diagonal
2026	**13.88%**	14.11%	Diagonal
Mean across three seeds	14.25%	**14.23%**	Effectively tied
Sample standard deviation across seed means	0.34 percentage points	0.39 percentage points	—

SNR	Diagonal	Matched dense	Difference: dense − diagonal
−10 dB	20.64%	**18.98%**	−1.66 pp
−5 dB	16.50%	**16.36%**	−0.14 pp
0 dB	14.03%	**13.97%**	−0.06 pp
+5 dB	**12.05%**	12.31%	+0.26 pp

SNR	Diagonal	Matched dense	Difference: dense − diagonal
+10 dB	**11.06%**	11.69%	+0.62 pp
+15 dB	**11.21%**	12.08%	+0.87 pp

At −10 dB, matched dense beat diagonal in all three seeds, suggesting a possible advantage in difficult noise. At +10 and +15 dB, diagonal beat matched dense in all three seeds. These are **pilot observations**, not evidence of statistical significance or proof that recurrence alone caused the differences.

Stage 5

Now, stages 3–4 trained sVAD using weighted cross-entropy (CE) with attention reconstruction weight λ = 0. Stage 5 asked whether including a clean-reference feature reconstruction objective improves noisy speech detection, and whether stopping reconstruction-loss gradients through the clean-reference branch changes performance or training dynamics.

Note: Clean reference inputs are the original dataset recordings, not guaranteed noise-free

Two gradient approaches:

- **Joint:** reconstruction-loss gradients affect the clean and noisy encoder paths.
- **Detached:** clean features are detached for reconstruction loss, so its gradients do not flow through the clean branch. This does not globally freeze encoder parameters.

A 60-epoch, multiple SNR sweeps were done on seed 42 to expand our observations. The results of our experiments were as follows:

Configuration	−10	−5	0	+5	+10	+15
A: baseline λ=0	**21.75**	17.37	13.81	12.81	12.33	13.23
D: detached λ=0.01	22.63	17.65	13.69	11.46	10.71	11.27
B: detached λ=0.1	22.22	18.05	13.66	11.59	11.38	11.54
E: detached λ=0.5	23.14	17.72	**13.07**	**11.13**	11.10	11.91
F: joint λ=0.01	22.71	17.73	13.30	11.52	11.18	12.03
C: joint λ=0.1	21.97	**17.10**	13.75	11.31	**10.93**	11.96
G: joint λ=0.5	22.63	18.38	14.88	12.92	11.82	**11.20**

Summarizing,

ID	λ	Clean-target gradient	Saved epoch	Mean validation HTER
A	0	Joint (irrelevant at λ=0)	60	15.216%
D	0.01	Detached	51	14.569%
B	0.1	Detached	57	14.741%
E	0.5	Detached	49	14.680%
F	0.01	Joint	60	14.745%
C	0.1	Joint	59	**14.502%**
G	0.5	Joint	56	15.304%

As visible, the best seed-42 validation score was joint λ=0.1; the detached configuration showed less sensitivity to the tested λ values.

All nonzero-λ candidates had **higher HTER at −10 dB** than the λ=0 baseline in this seed-42 validation sweep. Gains in six-SNR means therefore largely reflect other SNR ranges.

The above experiment was conducted on three seeds (42, 123 and 2026) to check repeatibility.

Configuration	Seed 42 HTER	Seed 123 HTER	Seed 2026 HTER	Three-seed mean	Sample SD
Baseline λ=0	15.22%	14.33%	14.62%	14.72%	0.45 pp
Detached λ=0.01	14.57%	14.78%	14.68%	**14.67%**	**0.10 pp**
Joint λ=0.1	14.50%	14.51%	15.16%	14.72%	0.38 pp

This stage was a bit of a wash. **No reproducible classification accuracy advantage was established** for the tested clean-reference MSE loss. The detached approach's ~0.05 pp mean advantage is small compared with between-seed variability. There are reasons for the same such as the fact that

- Gaussian noise corruption does not emulate all real noise environemnts
- Only three seeds were used for final comparison, no high confidence statistical significance claim.
- The parameter count alone does not establish a faithful reconstruction of the original SNN architecture.

At the end of Stage 5, we decided to proceed with **λ=0.0**, meaning **CE-only training**.

- **Lambda:** λ=0.0
- **Mode:** Joint (default, but irrelevant when λ=0)
- **Reason:** Neither detached nor joint MSE training demonstrated a consistent improvement in HTER across three random seeds.

Although detached training with λ=0.01 showed lower variability, its average accuracy improvement was negligible.

This proved to be a practical roadblock of this reconstruction.

Stage 6

Stage 6 set out to answer whether the spiking attention module improve frame-level voice activity detection in our reconstructed sVAD network relative to directly passing the encoder features to the recurrent classifier?

Since stage 5 showed that auxiallary clean-feature reconstruction loss difd not show a dependable accuracy benefit across three seeds, we proceeded training for this stage with λ = 0

We compared two models:

Component	Full attention	Attention bypass
SincNet + spiking-convolution encoder	Present	Present
Learned spiking attention mask	Present	Removed/bypassed
Feature passed to recurrent classifier	Attended noisy features $\hat{S} = Y \odot M$	Encoder features $\hat{S} = Y$
Diagonal spiking recurrent classifier	Present	Present
Encoder parameters	1,260	1,260
Attention parameters	2,283	0
Classifier parameters	770	770
Total parameters	**4,313**	**2,030**

We proceeded to run a 60 epochs per experiment; independent fresh initialization per run; seeds **42, 123, 2026**; batch size **8**; audio segment length **1.5 s**; training and checkpoint-selection validation SNRs **−10, −5, 0, +5, +10, +15.** The results were as follows:

Seed	Full checkpoint best epoch	Full HTER	Bypass checkpoint best epoch	Bypass HTER	Bypass − full (pp)
42	60	14.731%	51	14.661%	−0.069
123	55	14.393%	56	14.764%	+0.372
2026	60	14.864%	50	15.452%	+0.588
Mean	—	**14.662%**	—	**14.959%**	**+0.297**
Sample SD	—	**0.243 pp**	—	**0.430 pp**	—

Positive values in the final column favor learned attention. The best checkpoint's epoch is not necessarily the final training epoch, since checkpoint selection uses validation performance. The full model outperformed bypass in **2 of 3 seeds**, with a **0.297-percentage-point mean improvement** (about **2.0% relative** to the bypass mean HTER). Seed 42 slightly favored bypass.

SNR	Full HTER	Bypass HTER	Bypass − full (pp)
−10 dB	22.54%	22.95%	+0.41
−5 dB	17.33%	17.79%	+0.45
0 dB	13.31%	13.56%	+0.25
+5 dB	11.28%	11.80%	+0.52
+10 dB	11.17%	11.42%	+0.25
+15 dB	12.35%	12.25%	−0.10

Full attention performed better at **5 of 6** SNRs. It improved the noisiest cases by 0.41 pp (−10 dB) and 0.45 pp (−5 dB), but the largest observed improvement occurred at +5 dB (0.52 pp).

Stage 7

However, the issue was that we'd devoted only 2030 parameters to the bypass model, in comparison to the full attention model with 4313 parameters. Instead, the question then arose for Stage 7 was is a dense model, with attention bypass better? We explore the following architectures

Arm	Attention mode	Recurrent mode	Hidden neurons	Total parameters
A (full)	full	diagonal	32	4,313
B (lightweight reference)	bypass	diagonal	32	2,030
C (capacity-matched control)	bypass	dense	45	4,322

The results of the runs were as follows:

Architecture	Seed 42	Seed 123	Seed 2026	Three-seed mean	Sample SD (pp)
Full attention	14.125	13.390	14.230	**13.915**	**0.458**
Light bypass	14.133	13.428	15.387	14.316	0.992
Matched bypass	13.803	14.867	14.813	14.494	0.599

SNR	Full attention	Light bypass	Matched bypass	Lowest HTER
−10 dB	20.06	20.84	**18.90**	Matched bypass
−5 dB	**16.12**	16.64	16.36	Full
0 dB	**13.35**	13.97	14.61	Full
+5 dB	**11.65**	12.03	12.81	Full
+10 dB	**11.01**	11.05	12.06	Full
+15 dB	**11.29**	11.37	12.23	Full

Thus, the full attention diagonal model performs better than both the other models on 5/6 SNR values. It is considered to be our winner!

Final Notes

Previously recorded figures for the paper's QUT-NOISE-TIMIT evaluation are low-noise 4.0%, medium-noise 11.9%, and high-noise 19.1% HTER. These values **must not** be presented as directly comparable to our numbers: the current work uses a Speech Commands pilot, generated Gaussian noise and estimated speech labels, rather than the authors' original dataset and environmental-noise protocol. Architectural details of the original implementation also remain uncertain. No Loihi hardware power measurement or direct hardware-energy validation has been performed.

Reasons for the same are:

- Small pilot and three random seeds.
- Pseudo-ground-truth speech boundaries and potential label noise.
- Synthetic Gaussian noise instead of the paper's environmental noise distributions.
- Exposure of the test split earlier in the project; not an untouched holdout.
- Comparison confounded by changes to recurrent connectivity/width in capacity matching.
- SNR-dependent FAR behavior remains unexplained.
- No hardware power, latency or energy-per-inference measurements.
- Not a bit-for-bit reproduction of the published system.

